Gaurav Jain

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Research Interests

Human-computer interaction (HCI), computer vision, assistive technologies, robotics, deep learning

Education

2020–Present **Columbia University, Graduate School of Arts and Sciences**, New York, NY
Ph.D. in Computer Science | GPA: 4.13/4.00
Specialization: Human-Computer Interaction
Advisor: Dr. Brian A. Smith
Featured Coursework: User Interface Design, Human-Computer Interaction, Computational Aspects of Robotics

2016–2020 **Delhi Technological University**, New Delhi, India
B.Tech in Computer Science | GPA: 9.38/10.0 *Featured Coursework*: Computer Vision, Soft Computing, Machine Learning, Artificial Intelligence, Digital Image Processing, Swarm Intelligence, Distributed Systems.

Research Experience

2020–Present **Columbia University**, New York, NY
Graduate Research Assistant, Computer-Enabled Abilities Lab (CEAL)
- **Spotio**: A wearable camera system for blind and visually impaired people to create personalised maps of indoor spaces and navigate independently (GitHub).

Frameworks: Robot Operating System (ROS), Swift, Unity, Python

2020 **Université Clermont Auvergne**, Clermont-Ferrand, France
Summer Research Intern, Endoscopy and Computer Vision (EnCoV)
- Patient-specific organ tracking in laparoscopic images by deep learning (GitHub).

Frameworks: Blender, Gmsh, Elmer, Paraview, Python (PyTorch)

2019–20 **Delhi Technological University**, New Delhi, India
Undergraduate Research Assistant, Machine Learning Research Lab
- Designed and implemented a Transformer-based deep neural network architecture for sketch recognition. Published paper at ECAI 2020 (Paper), and IEEE Trans. CDS (Paper).

Frameworks: Python (TensorFlow)

2018–20 **Indian Institute of Technology (IIT)**, New Delhi, India
Research Intern, School of Information Technology
- Developed a deep learning based breast cancer detection model for scale invariant detection of masses and calcifications. Supported by All India Institute of Medical Sciences (AIIMS).

Frameworks: Python (TensorFlow), MATLAB

2019 **Indian Institute of Science (IISc)**, Bangalore, India
Summer Research Fellow, Department of Computational and Data Sciences
- Deep Neural-Network Based Sinogram Super-resolution and Bandwidth Enhancement for Limited Data Photoacoustic Tomography. Paper published in IEEE Transactions (Paper).

Frameworks: Python (PyTorch), MATLAB (k-Wave Toolbox)

2018–19 **Defense Research & Development Organisation (DRDO)**, New Delhi, India
Research Assistant, Scientific Analysis Group

- Designed a graph based fusion approach for a multimodal biometric system that fuses fingerprint, face and iris scans in a highly secure and cancelable manner. Paper published in IEEE Transactions (Paper).

Frameworks: MATLAB

2018 **National University of Singapore (NUS)**, Singapore
Academic Intern (Summer), School of Computing

- Training program on Artificial Neural Networks & Big Data Analytics. Batch Topper of the Quiz & Project Presentation.

Frameworks: Python, R, Hadoop, Cloudera, Hive

Publications

2020 **Gaurav Jain***, Shivang Chopra*, Suransh Chopra*, Anil Singh Parihar. "*Attention-Net: An Ensemble Sketch Recognition Approach using Vector Images*" Published in IEEE Transactions on Cognitive and Developmental Systems, 2020.
PDF • DOI

2020 **Gaurav Jain***, Shivang Chopra*, Suransh Chopra*, Anil Singh Parihar. "*TransSketchNet: attention-based Sketch Recognition using Transformers*" (Highlight Paper) To appear at 24th European Conference on Artificial Intelligence, ECAI 2020.
PDF • DOI

2020 Navchetan Awasthi*, **Gaurav Jain***, S. K. Kalva, Manojit Pramanik, Phaneendra K. Yalavarthy. "*Deep Neural-Network Based Sinogram Super-resolution and Bandwidth Enhancement for Limited Data Photoacoustic Tomography*" Published in IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control (Special issue on Deep Learning in Medical Ultrasound), 2020 (in press).
PDF • DOI • CODE

2019 Gurjit Singh Walia, **Gaurav Jain**, Nipun Bansal, Kuldeep Singh. "*Adaptive Weighted Graph Approach to Generate Multimodal Cancelable Biometric Templates*" Published in IEEE Transactions on Information Forensics and Security, vol. 15, pp. 1945-1958, 2020.
PDF • DOI

** Indicates Equal Contribution*

Awards & Honors

2020 **Greenwoods Fellowship**, *Department of Computer Science, Columbia University*
Received funding of $\$15,400 +$ tuition fee for the fall semester (2020).

2019 **Summer Research Fellowship**, *Indian Academy of Sciences, Govt. of India*
Received funding for summer research internship at the Indian Institute of Science, Bangalore.

2016 **Joint Entrance Examination (JEE MAIN'16)**, *Central Board of Secondary Education*
All India Rank 3366 among approx 1.5 million candidates.

Teaching Experience

2021–Present **Teaching Assistant**, Columbia University
Graduate Level Courses:

- COMS E6998: Human-Computer Interaction (Spring 2021)
  Instructor: Prof. Brian A. Smith

Leadership & Professional Services

2017–2020 **Grad Application Mentor**, Department of Computer Science, Columbia University
Volunteer, Pre-Submission Application Review Program (2020)

- Reviewed PhD application material for students with less access to research mentoring.

2017–2020 **Optima, Machine Learning Society**, Delhi Technological University
Co-Founder & President (2019–20)

- Organized peer-taught lessons, curated lecture series, and moderated online discussions.

Mentoring & Advising

2021–Present **Research Project Students**, Columbia University

- Yuanyang Teng (M.S.; Spring 2021)
- Jessica Peng (B.S.; Spring 2021)
- David Cho (M.S.; Spring 2021)
- Yunhao Xing (M.S.; Spring 2021)

Skills

Proficient with C, C++, Python (TensorFlow, PyTorch, Keras, OpenCV), MATLAB, LaTeX, Linux, ROS, Swift, Unity, Balsamiq, Figma. *Familiar* with R, Hive, Cloudera, Docker, Blender, Paraview, HTML, CSS, Javascript.

References

Brian A. Smith
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Defense Research & Development Organization (DRDO)
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Computational & Data Sciences
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